

Week 01: Information Theory and Networking

This week explores how information is quantified and represented in computer systems. We don't treat data as raw bits instead, we analyze how much actual information is present using Shannon entropy and how representation choices impact efficiency.

1) Information vs Data

- Information → reduction in uncertainty (measured by entropy)
- Data → representation of that information (measured in bits/bytes)

Example:

- English text 1.5 bits/character
- ASCII uses 8 bits/character

Efficiency = $1.5 / 8 = 19\%$

2) Shannon Entropy:

Measures average information per symbol

Lower entropy → more predictable → easier to compress

Higher entropy → more randomness → harder to compress

Examples:

- Fair coin → $H = 1$ bit
- Biased coin (0.9 / 0.1) → $H \approx 0.47$ bits

Less uncertainty = less information

3) Units Matter :

Network speeds → SI (base 10)

- 1 Mb/s = 1,000,000 bits/sec

Memory/storage → Binary (base 2)

- 1 MiB = 1,048,576 bytes

4) Encoding Tradeoffs

ASCII

- Fixed-length (7–8 bits)
- Efficient for English only

UTF-8

- Variable-length (1–4 bytes)
- Supports global text
- Tradeoff: flexibility vs size

Base64

- Converts binary → text
- Adds 33% overhead

5) Binary Formats vs Human-Readable Formats :

- Json: Readable and flexible , easily integrated however it has a large size.
- Binary : Binary is compact and efficient but it's hard to work with.

Therefore in Networking we are constantly balancing: Efficiency vs Readability vs Compatibility

6)Multimedia Data Rate Why do we need compression?

Audio (CD quality): $44.1 \text{ kHz} \times 16 \text{ bits} \times 2 \text{ channels} = 1.4 \text{ Mb/s}$

Video (1080p raw) = 1.5 Gb/s

Therefore without compression, streaming would be impossible.

7) Network connection: All of the concepts above directly impact Bandwidth requirements and the Protocol design as well as Transport choices TCP or UDP

Personal insight: Networks don't transmit "data" they transmit representations of information, and the efficiency of those representations determines whether systems are feasible at scale. I found entropy and reducing uncertainty to be the most interesting part of the lectures; it's something that I have not been introduced to before (Learned Entropy from biology only) .